LoGlo-CLIP: Local-to-Global Weighted Layer Fusion for CLIP Image-Text Retrieval

22200376 Myeongjin Seon

Abstract

Vision-language pretraining models align images and text in a shared embedding space and have shown strong performance across multimodal tasks such as image-text retrieval and zero-shot classification. Among them, CLIP learns image-text alignment by applying a contrastive objective to large-scale image-text pairs. However, CLIP-based retrieval typically relies only on the final image embedding, leaving the intermediate-layer representations of the Vision Transformer largely unused. Since the representations at different layers of a ViT tend to differ in character, combining several layers offers room to improve retrieval performance.

In this work, we propose LoGlo-CLIP, a framework that makes use of the intermediate-layer representations of a frozen CLIP ViT-B/32 image encoder. The framework extracts the CLS token from layers 3, 6, 9, and 12 and combines them through a static learnable weighted sum. The fused feature is projected into the CLIP embedding space and merged with the original CLIP embedding through a residual connection. To examine how fusion complexity affects retrieval, we additionally compare this with self-attention and cross-attention variants. On Flickr30k image-text retrieval, all three strategies improve over the CLIP baseline. Among the two attention-based variants, which share the same number of trainable parameters, cross-attention achieves higher image-to-text performance; meanwhile, the simple weighted sum, using approximately seven times fewer parameters, attains the best text-to-image performance. These results suggest that the most effective fusion depends on the retrieval direction, and that the added cost of attention is not always justified. We further analyse the design through layer and feature-type ablations, and examine model behaviour through hard-negative retrieval, robustness evaluation, and visualization.

1 Introduction

Image-text retrieval is a multimodal task that aligns images and their natural-language descriptions in a shared embedding space, allowing a caption to be retrieved for a given image or an image for a given caption. Vision-language pretrained models such as CLIP, trained on large-scale image-text pairs with a contrastive objective, have achieved strong image-text alignment.

In CLIP-based retrieval, however, the final image embedding is typically used on its own. A Vision Transformer image encoder builds up its visual representation gradually across successive Transformer layers, and the representations at different layers tend to differ in character. Relying on the final layer alone may therefore underuse the visual information contained in the intermediate layers.

To address this, we propose LoGlo-CLIP, which extracts the CLS token from layers 3, 6, 9, and 12 of the CLIP ViT-B/32 image encoder and combines them. The CLIP backbone is kept frozen, and only a small number of fusion parameters and a projector are trained, making the method parameter-efficient.

The main contributions of this work are as follows. First, we propose a framework that combines the multi-layer CLS representations of a frozen CLIP ViT for image-text retrieval. Second, we compare the proposed weighted-sum fusion with two attention-based variants of increasing complexity — self-attention and cross-attention — to examine how the fusion mechanism affects retrieval performance on Flickr30k. Third, we analyse the effectiveness and limitations of the proposed design through layer ablation, patch-mean ablation, hard-negative retrieval, robustness evaluation, and visualization.

2 Related Work

2.1 Vision-Language Pretraining

Vision-language pretraining aims to align images and text in a shared embedding space so that the learned representations transfer to a range of multimodal tasks. CLIP [Radford et al., 2021] learns this alignment by applying a contrastive objective to large-scale image-text pairs. Building on this paradigm, ALIGN [Jia et al., 2021] scales it further using a large and noisy corpus of alt-text collected without an expensive filtering step, while BLIP [Li et al., 2022] bootstraps captions to make better use of noisy web data and to address both understanding and generation. Through this line of work, contrastive vision-language pretraining has become a central approach to multimodal representation learning.

2.2 CLIP

This work uses CLIP [Radford et al., 2021] as its retrieval backbone. CLIP adopts a dual-encoder design, in which a separate image encoder and text encoder map the two modalities into a shared embedding space, and a contrastive loss is used to align them. The resulting representations perform well on zero-shot classification and image-text retrieval, and transfer to a variety of downstream tasks without task-specific fine-tuning. In this study, the ViT-B/32 image encoder is kept frozen, and the focus is placed on making fuller use of its internal representations.

2.3 Vision Transformer Representation

The Vision Transformer [Dosovitskiy et al., 2021] divides an image into fixed-size patches, treats each patch as a token, and feeds the resulting sequence into a Transformer encoder. A useful property of this design is that the hidden state and patch tokens at each layer can be accessed directly. An analysis of internal ViT representations [Raghu et al., 2021] has shown that, unlike CNNs, ViTs form relatively uniform representations across layers, with residual connections strongly propagating lower-layer features to higher layers. Since the representations at different layers therefore differ in character, relying on the final layer alone may not fully exploit the information contained in the intermediate layers. Motivated by this observation, we combine the intermediate hidden states of the CLIP ViT-B/32 image encoder to build a multi-layer visual representation that does not depend on the final layer alone.

2.4 Image-Text Retrieval

Image-text retrieval is commonly divided into image-to-text and text-to-image retrieval. Recall@K is the metric most widely used for evaluation. In addition to Recall@K, this study reports MRR, Median Rank, Mean Rank, and nDCG@10 in order to assess ranking quality from several complementary perspectives.

2.5 Multi-layer Feature Fusion

Rather than relying on the output of a single layer, combining representations from several layers makes it possible to draw on information at different levels and to form a richer representation. Such combination can be designed in various ways, ranging from a simple weighted sum to attention-based dynamic fusion, and these choices differ in expressiveness and computational cost. Building on this idea, we compare an input-independent weighted sum with self-attention and cross-attention fusion under the same setting, and analyse the relationship between the complexity of the fusion mechanism and retrieval performance.

3 Proposed Method

3.1 Overall Architecture

The proposed LoGlo-CLIP adds a lightweight visual fusion adapter on top of a frozen CLIP ViT-B/32 image encoder. The overall architecture is illustrated in Figure 1. An input image is passed through the CLIP image encoder, and the CLS token is extracted from the hidden states of layers

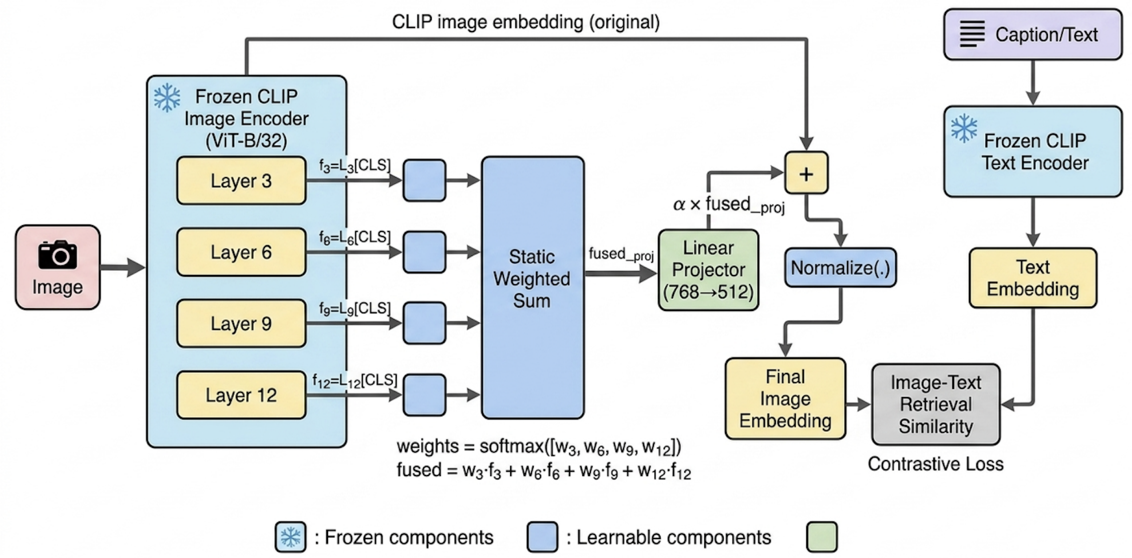

Figure 1: Overall architecture of LoGlo-CLIP. The CLS tokens from layers 3, 6, 9, and 12 of the frozen CLIP image encoder are combined by a static weighted sum, projected into the CLIP embedding space, and merged with the original CLIP image embedding through a residual connection. Only the fusion weights and the linear projector are trained.

3, 6, 9, and 12. These layer-wise CLS representations are combined into a single visual feature through a static weighted sum, mapped to the CLIP embedding dimension by a linear projector, and merged with the original CLIP image embedding through a residual connection. The resulting image embedding is then compared with the text embedding from the frozen CLIP text encoder using cosine similarity for retrieval. Throughout this process the CLIP backbone is kept frozen, and only the fusion weights and the projector are trained. This section describes the fusion mechanism used in LoGlo-CLIP; a comparison with alternative fusion strategies is presented in Section 4.2.

3.2 Layer-wise CLS Feature Extraction

We define the set of selected layers of the CLIP ViT-B/32 image encoder as $L = \{3, 6, 9, 12\}$. For each layer $l \in L$, the CLS token is taken from the corresponding hidden state and denoted by $h_l = H_l^{\text{CLS}}$, where $h_l \in \mathbb{R}^{768}$. This set was chosen so as to span the encoder evenly from lower to higher layers. Because the representations at different ViT layers differ in character, using the CLS tokens of several layers together, rather than a single layer, allows the model to capture information that the final layer alone does not fully reflect.

3.3 Static Weighted-Sum Fusion

LoGlo-CLIP combines the layer-wise CLS features using a set of learnable static weights. For each layer l, a scalar parameter a_l is defined and normalised with a softmax to obtain the layer weight w_l:

$$w_l = \frac{\exp(a_l)}{\sum_{j \in L} \exp(a_j)}. \tag{1}$$

The CLS representations of the selected layers are then combined by a weighted sum to form the fused feature f:

$$f = \sum_{l \in L} w_l \, h_l. \tag{2}$$

Because these weights are input-independent and shared across all images, the learned values of w_l can be inspected directly to see how much each layer contributes to retrieval. A further advantage of this design is that it is simple and uses few trainable parameters, which makes it well suited as a lightweight adapter on top of a frozen CLIP representation.

3.4 Projection and Residual Fusion

The fused feature f is passed through a linear projector W_p to match the dimension of the CLIP embedding:

$$z_{\text{fusion}} = W_p\, f. \tag{3}$$

To preserve the image-text alignment that CLIP acquires during pretraining, the fused feature is not used on its own but is combined with the original CLIP image embedding z_{clip} through a residual connection:

$$z_{\text{final}} = \text{Normalize}\big(z_{\text{clip}} + \alpha\, z_{\text{fusion}}\big). \tag{4}$$

Here α controls how strongly the fused feature is incorporated, and we set $\alpha = 0.3$ in this study. This residual design retains the original CLIP representation while adding the multi-layer feature in a complementary manner.

3.5 Feature Type Ablation: CLS vs. Patch Mean

To examine how the choice of feature representing each layer affects retrieval performance, we compare the CLS token with the patch mean. The default LoGlo-CLIP uses the CLS token of each layer, whereas the patch-mean variant replaces it with the average of the patch tokens at that layer:

$$h_l^{\text{patch}} = \frac{1}{N} \sum_{i=1}^{N} p_{l,i}, \tag{5}$$

where $p_{l,i}$ is the i-th patch token of layer l and N is the number of patch tokens. This comparison is intended to show how patch-level spatial information affects retrieval.

4 Experiments and Results

4.1 Setup

Dataset. All experiments were conducted on Flickr30k [Young et al., 2014], a widely used benchmark for image-text retrieval that pairs images with five natural-language captions each. The fusion adapter was trained on the train split, and the checkpoint with the lowest validation loss was selected using the validation split. Final retrieval performance is reported on the test split.

Implementation Details. All models use CLIP ViT-B/32 as the backbone. Both the image encoder and the text encoder were kept frozen throughout training; only the layer-wise fusion module and the linear projector were updated. The default layer combination was set to $\{3, 6, 9, 12\}$. Training was carried out for 5 epochs with a batch size of 64, a learning rate of 1×10^{-4}, a weight decay of 0.01, and a residual coefficient of $\alpha = 0.3$. The total number of trainable parameters is approximately 0.39M, comprising the fusion weights and the projector. This corresponds to a parameter-efficient adaptation setting in which a small adapter is placed on top of a frozen CLIP representation, rather than fine-tuning the entire backbone.

Evaluation Metrics. Image-to-text (I2T) and text-to-image (T2I) retrieval are both evaluated using Recall@1, Recall@5, and Recall@10 as the primary metrics. MRR, Median Rank, Mean Rank, and nDCG@10 are also reported to assess ranking quality in greater detail.

Compared Models. Three models are compared in the main experiments. The first is CLIP ViT-B/32, the baseline that uses the original frozen CLIP embeddings without modification. The second is CLIP+WeightedSum-CLS, the proposed method, which extracts the CLS token from layers $\{3, 6, 9, 12\}$, combines them with a static learnable weighted sum, and applies residual fusion. The third is CLIP+WeightedSum-PatchMean, used for the feature-type ablation, which replaces the CLS token with the mean of the patch tokens at each selected layer. For the fusion strategy comparison, two additional attention-based variants — Self-Attention (SA) and Cross-Attention (CA) — are also evaluated.

4.2 Main Results

Table 1 presents the retrieval results on the Flickr30k test set. The CLIP ViT-B/32 baseline achieves an I2T R@1 of 70.50 and a T2I R@1 of 68.10. The proposed LoGlo-CLIP (WeightedSum-CLS) improves over the baseline in both directions, reaching an I2T R@1 of 74.90 (+4.40) and a T2I R@1 of 75.00 (+6.90). The improvement is notably larger in the T2I direction, suggesting that combining intermediate-layer CLS representations produces an image embedding that is more discriminative for caption-to-image matching. These gains are obtained with approximately 0.39M trainable parameters on top of a frozen CLIP backbone, without updating the image or text encoder.

Model	Image-to-Text			Text-to-Image		
	R@1	R@5	R@10	R@1	R@5	R@10
CLIP ViT-B/32	70.50	91.00	95.40	68.10	89.80	94.20
CLIP+WeightedSum-CLS $\{3, 6, 9, 12\}$	**74.90**	**93.80**	**96.90**	**75.00**	**92.80**	**96.20**

Table 1: Main retrieval results on the Flickr30k test set. I2T = image-to-text; T2I = text-to-image.

Fusion Strategy Comparison. Table 2 compares the three fusion strategies. The two attention-based variants, SA and CA, each use approximately 2.75M trainable parameters, whereas WS uses 0.39M. In the I2T direction, performance increases with complexity: CA (76.60) > SA (74.20) > WS (74.90). In the T2I direction, however, WS (75.00) outperforms both CA (74.00) and SA (73.70), showing that the simpler method achieves the best performance in that direction. These results suggest that the most effective fusion strategy depends on the retrieval direction, and that the additional parameter cost of attention is not always justified. It is also worth noting that SA and CA share identical parameter counts yet differ by 2.4 points in I2T R@1, indicating that the performance differences arise from the fusion mechanism itself rather than from the increase in parameter count.

Method	Params	I2T		T2I	
		R@1	R@5	R@1	R@5
CLIP ViT-B/32 (baseline)	—	70.50	91.00	68.10	89.80
WeightedSum (WS)	0.39M	74.90	**93.80**	**75.00**	**92.80**
Self-Attention (SA)	2.75M	74.20	93.20	73.70	92.40
Cross-Attention (CA)	2.75M	**76.60**	93.80	74.00	92.80

Table 2: Fusion strategy comparison on Flickr30k. All variants use layers $\{3, 6, 9, 12\}$. Trainable parameters exclude the frozen CLIP backbone.

4.3 Ablation Study

Layer Combination. Table 3 shows the effect of varying the layer combination. Using the final layer alone ($\{12\}$) already forms a strong baseline. Including intermediate layers tends to improve performance on several metrics: $\{3, 6, 9, 12\}$ achieves the highest I2T R@1 (74.90), while $\{6, 9, 12\}$ attains the highest T2I R@1 (75.40). $\{3, 6, 9, 12\}$ maintains competitive performance in both directions and is therefore used as the default setting.

Layers	Val. Loss	I2T			T2I		
		R@1	R@5	R@10	R@1	R@5	R@10
$\{12\}$	0.2295	74.00	93.40	**97.00**	74.90	**93.00**	96.20
$\{6, 9, 12\}$	0.2265	74.30	93.50	96.60	**75.40**	92.50	96.00
$\{3, 6, 9, 12\}$	0.2279	**74.90**	**93.80**	96.90	75.00	92.80	**96.20**

Table 3: Layer combination ablation for WeightedSum fusion on Flickr30k.

Feature Type: CLS vs. Patch Mean. Table 4 compares the CLS token with the patch mean as the per-layer feature. PatchMean achieves a slightly higher I2T R@1 (73.90 vs. 74.90) and MRR (0.8237 vs. 0.8203). In the T2I direction, however, it falls considerably short of CLS across all metrics, with a T2I R@1 of 71.70 (vs. 75.00) and a T2I MRR of 0.8033 (vs. 0.8271). This indicates that while patch-level features offer some benefit for I2T retrieval, the CLS token aligns more consistently with CLIP's global image-text representations in the T2I direction. Taken together, CLS yields more balanced performance across both directions and is therefore adopted as the default setting.

Variant	Image-to-Text				Text-to-Image			
	R@1	R@5	R@10	MRR	R@1	R@5	R@10	MRR
WeightedSum-CLS	**74.90**	**93.80**	96.90	0.8203	**75.00**	**92.80**	**96.20**	**0.8271**
WeightedSum-PatchMean	73.90	93.00	**97.00**	**0.8237**	71.70	90.80	94.90	0.8033

Table 4: Feature type ablation (CLS token vs. patch mean) with layers $\{3, 6, 9, 12\}$ on Flickr30k.

5 Discussion

5.1 Hard Negative Retrieval

Standard retrieval evaluates a model's ability to retrieve the correct caption from the full caption pool. Hard-negative retrieval provides a complementary assessment by constructing, for each image, a candidate set that pairs the positive caption with semantically similar negative captions selected on the basis of CLIP text embedding similarity. The model is then evaluated on whether it assigns the highest score to the positive caption.

As shown in Table 5, the proposed LoGlo-CLIP outperforms the CLIP baseline on Top-1 Accuracy (88.20 vs. 87.20), MRR (0.9344 vs. 0.9293), and Mean Margin (0.0452 vs. 0.0405). These results indicate that combining multi-layer CLS representations improves the model's ability to discriminate between semantically similar captions.

Model	Top-1 Acc	MRR	Mean Rank	Median Rank	Mean Margin
CLIP ViT-B/32	87.20	0.9293	1.177	1.000	0.0405
CLIP+WeightedSum $\{3, 6, 9, 12\}$	**88.20**	**0.9344**	**1.164**	1.000	**0.0452**

Table 5: Hard-negative retrieval results on Flickr30k.

5.2 Robustness Evaluation

Table 6 reports I2T retrieval performance under five image perturbations: Gaussian blur, grayscale conversion, colour jitter, centre crop, and occlusion. LoGlo-CLIP maintains a higher absolute R@1 than the CLIP baseline under most perturbation conditions. Under colour jitter and centre crop, it also shows a smaller R@1 drop (4.00 and 2.70, respectively) than the baseline (4.50 and 3.80), indicating greater robustness to those corruptions.

Under grayscale and occlusion, however, the R@1 drop for LoGlo-CLIP (14.80 and 17.90) is larger than for the baseline (10.50 and 15.90). In both cases the absolute R@1 values remain similar between the two models (grayscale: 59.40 vs. 60.00; occlusion: 56.30 vs. 54.60), and performance reversal occurs only in the grayscale condition. These results suggest that LoGlo-CLIP cannot be considered uniformly more robust across all corruptions, although it achieves higher absolute retrieval performance under most conditions.

Model	Corruption	R@1	R@5	R@10	MRR	R@1 Drop
CLIP ViT-B/32	Original	70.50	91.00	95.40	0.7961	0.00
	Gaussian blur	65.50	86.70	92.30	0.7507	5.00
	Grayscale	60.00	83.20	89.50	0.7012	10.50
	Colour jitter	66.00	88.20	93.70	0.7596	4.50
	Centre crop	66.70	88.90	94.00	0.7679	3.80
	Occlusion	54.60	80.40	86.00	0.6579	15.90
CLIP+WeightedSum $\{3, 6, 9, 12\}$	Original	74.20	93.50	96.70	0.8256	0.00
	Gaussian blur	67.90	90.00	94.30	0.7756	6.30
	Grayscale	59.40	82.60	89.40	0.7000	14.80
	Colour jitter	70.20	90.60	94.40	0.7936	4.00
	Centre crop	71.50	91.80	94.90	0.8028	2.70
	Occlusion	56.30	80.80	87.20	0.6744	17.90

Table 6: Robustness evaluation: I2T retrieval under image perturbations on Flickr30k. R@1 Drop is relative to each model's clean performance.

5.3 Visualization and Layer Weight Analysis

Figure 2 shows the learned softmax weights assigned to each selected layer after training. Layer 12 receives a slightly higher weight ($w_{12} \approx 0.29$) than the other three layers (each ≈ 0.24), indicating that the final semantic layer contributes most to the fused representation. At the same time, the intermediate layers 3, 6, and 9 each retain non-negligible weights, suggesting that they provide complementary information that the final layer alone does not capture. This is consistent with the retrieval improvements observed when moving from the single-layer $\{12\}$ setting to the full $\{3, 6, 9, 12\}$ combination in the layer ablation (Table 3).

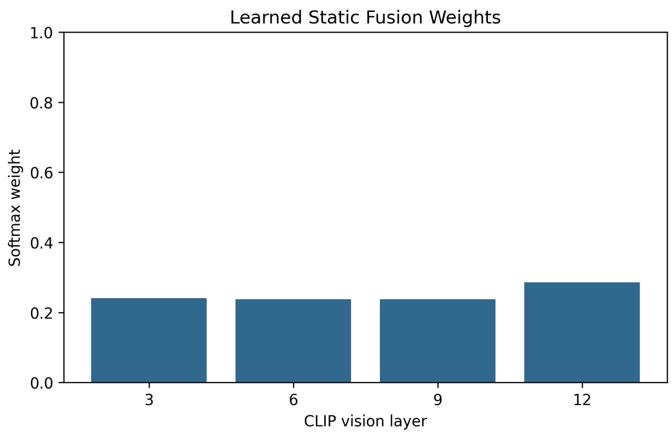

Figure 2: Learned static fusion weights (w_l) for each selected layer after training on Flickr30k. Layer 12 receives a slightly higher weight, while intermediate layers retain non-negligible contributions.

Figure 3 visualises the CLS relevance maps at layers 3, 6, 9, and 12, alongside the CLIP final representation, the LoGlo weighted-sum fusion representation, and the cumulative attention rollout. As the layer depth increases, the attention progressively shifts from broader regions towards semantically relevant areas such as the people in the scene. The LoGlo fusion representation combines these layer-wise patterns, producing an attention map that reflects both local and global visual information. The rollout path further illustrates the cumulative propagation of attention across layers.

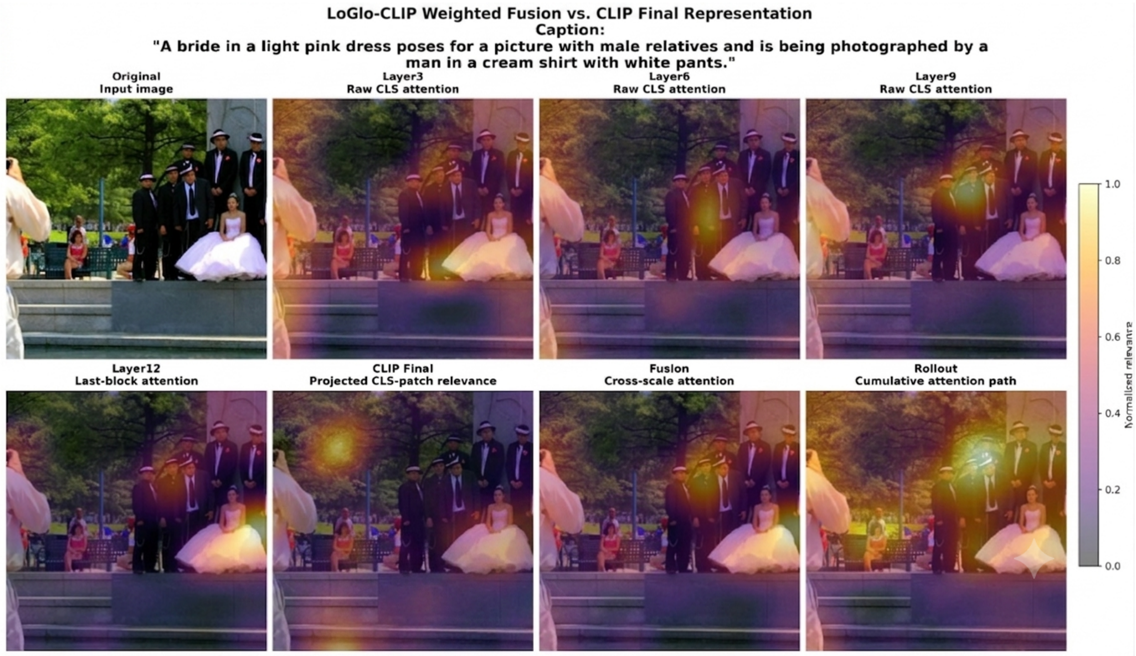

Figure 3: Layer-wise CLS relevance maps for a sample image from Flickr30k. Top row (left to right): original image, layer 3, layer 6, and layer 9 CLS relevance. Bottom row: layer 12, CLIP final representation, LoGlo weighted-sum fusion representation, and cumulative attention rollout. Warmer colours indicate higher normalised relevance.

6 Conclusion

This paper proposed LoGlo-CLIP, a lightweight framework that combines the intermediate-layer CLS representations of a frozen CLIP ViT-B/32 image encoder for image-text retrieval. The CLS tokens from layers 3, 6, 9, and 12 are combined through a learnable weighted sum, projected into the CLIP embedding space, and merged with the original CLIP image embedding via a residual connection.

Experiments on Flickr30k show that the proposed WeightedSum-CLS improves retrieval performance over the CLIP baseline in both directions. A comparison of three fusion strategies reveals that the optimal strategy depends on the retrieval direction: while cross-attention achieves the highest image-to-text performance, the simpler weighted sum, using approximately seven times fewer parameters, attains the best text-to-image performance. Layer and feature-type ablations confirm the contribution of intermediate layers and the advantage of CLS-based representations for balanced retrieval. Additional analysis through hard-negative retrieval and robustness evaluation shows that the proposed method improves fine-grained caption discrimination and maintains higher absolute retrieval performance under most image corruptions, although its robustness advantage is not consistent across all perturbation types.

Future work could explore image-dependent adaptive weighting and corruption-aware training to further improve the robustness of the proposed fusion design.

References

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc V. Le, Yun-Hsuan Sung, Zhen Li, and Tom Duerig. Scaling up visual and vision-language representation learning with noisy text supervision. In *Proceedings of the 38th International Conference on Machine*

Learning (ICML), volume 139 of *Proceedings of Machine Learning Research*, pages 4904–4916, 2021.

Junnan Li, Dongxu Li, Caiming Xiong, and Steven C. H. Hoi. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, volume 162 of *Proceedings of Machine Learning Research*, pages 12888–12900, 2022.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763, 2021.

Maithra Raghu, Thomas Unterthiner, Simon Kornblith, Chiyuan Zhang, and Alexey Dosovitskiy. Do vision transformers see like convolutional neural networks? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 12116–12128, 2021.

Peter Young, Alice Lai, Micah Hodosh, and Julia Hockenmaier. From image descriptions to visual denotations: New similarity metrics for semantic inference over event descriptions. *Transactions of the Association for Computational Linguistics*, 2:67–78, 2014.